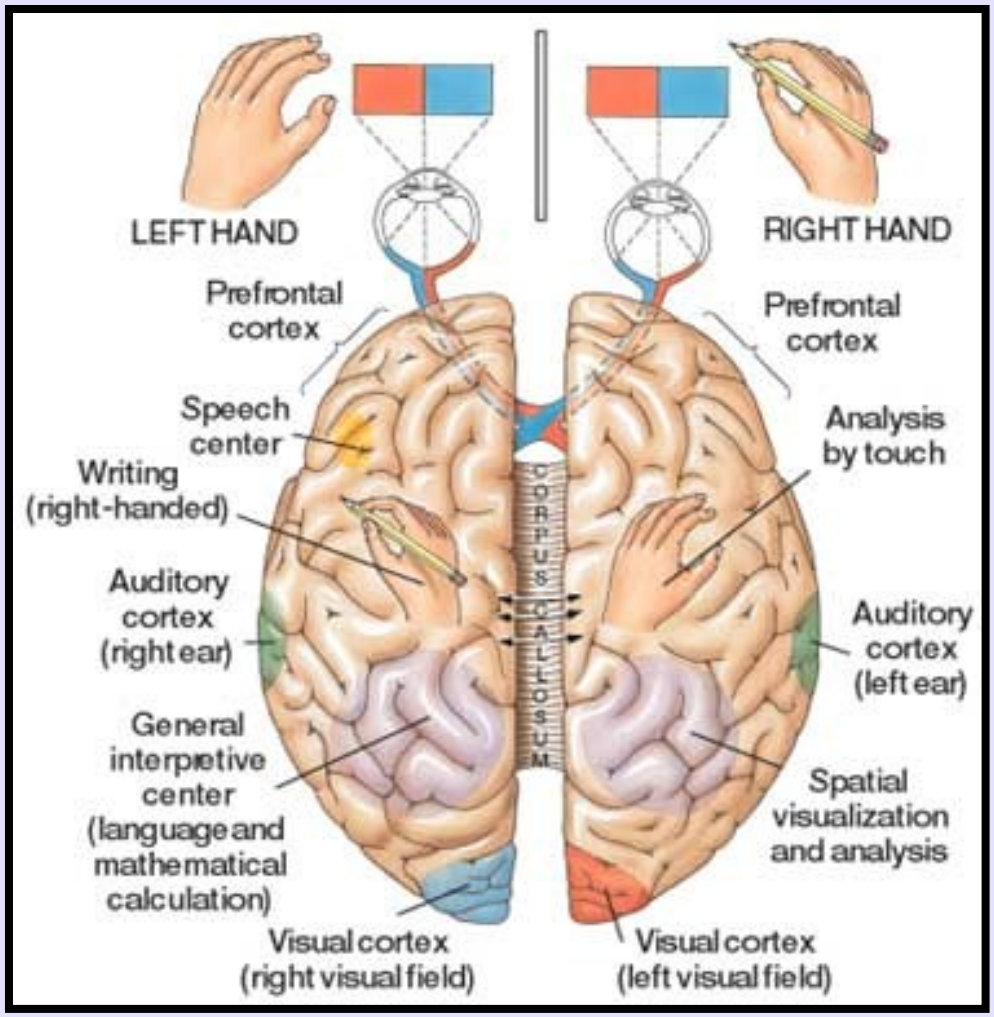
SURFACES, LOBES, SULCI, GYRI OF CEREBRAL HEMISPHERES

LEARNING OBJECTIVES

- At the end of the lecture the student should be able to know:
- Surfaces of cerebral hemispheres.
- Lobes of cerebral hemispheres
- Sulci and gyri of cerebral hemispheres

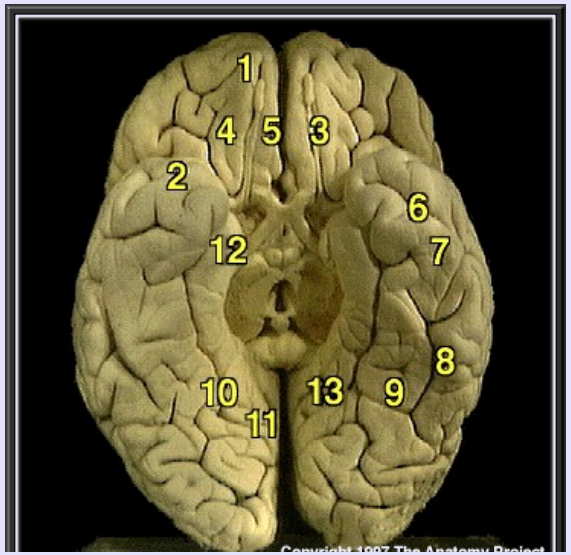
CEREBRAL HEMISPHERES

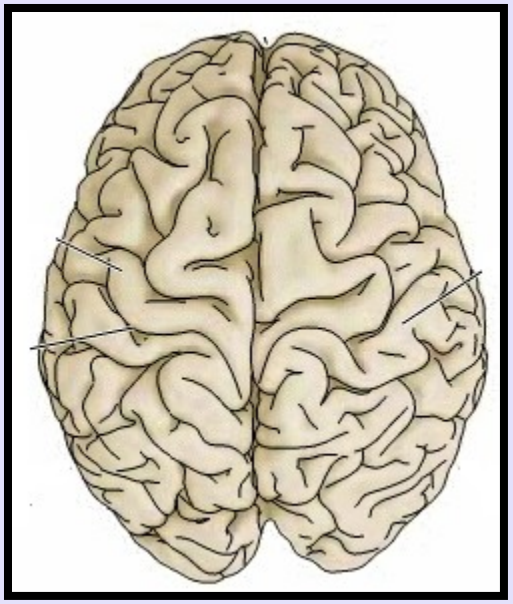
- The cerebral hemispheres constitute the largest part of the brain
- Divided into right and left hemispheres.
- The hemispheres are separated medially by a deep cleft, named the **longitudinal cerebral fissure**
- Linked by corpus callosum.

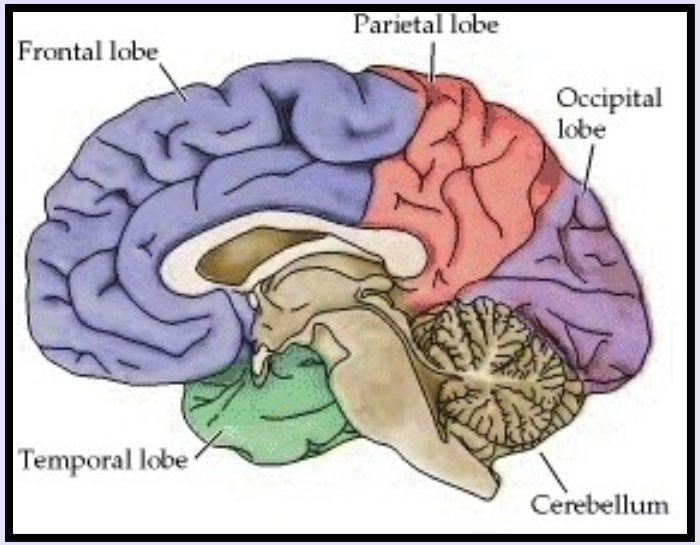


Surfaces of cerebral hemispheres

- Lateral surface:
  - convex
- Medial surface
  - flat and vertical
  - Separated by longitudinal fissure
- Inferior surface
  - Anterior = orbital surface of the frontal lobe
  - Middle = under surface of the temporal lobe
  - Posterior = **tentorial surface**

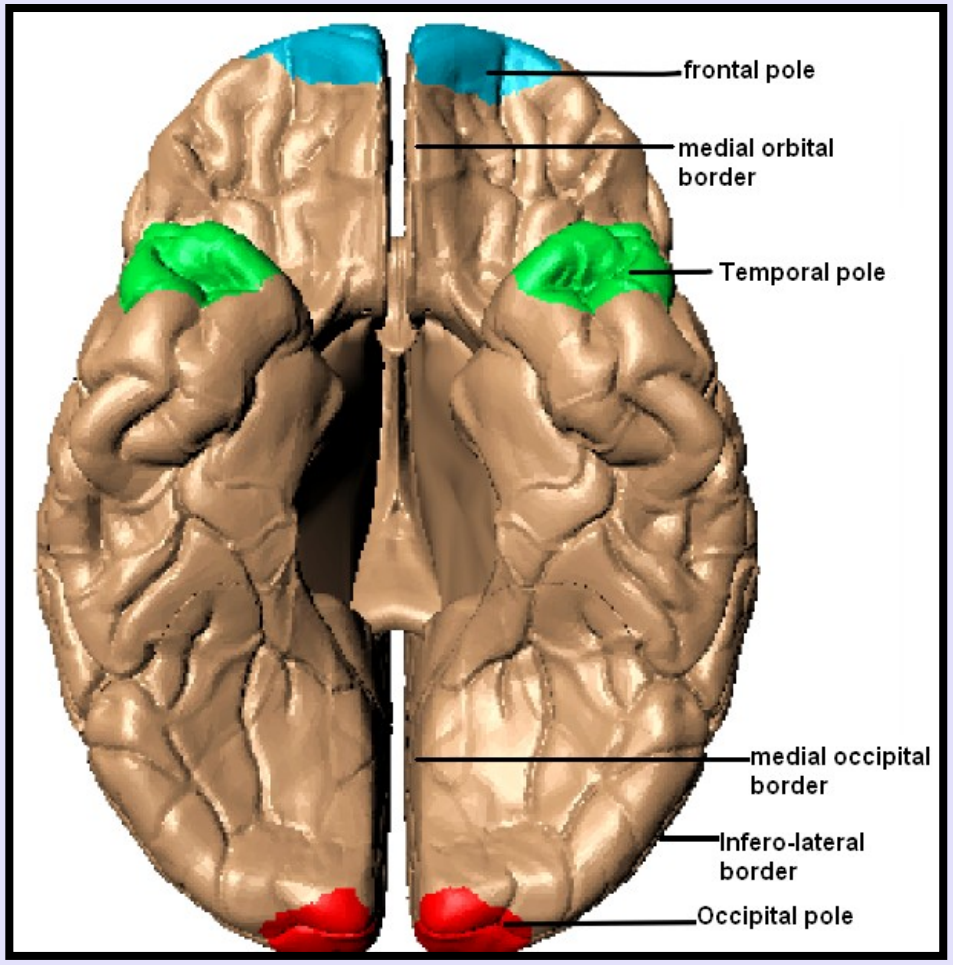






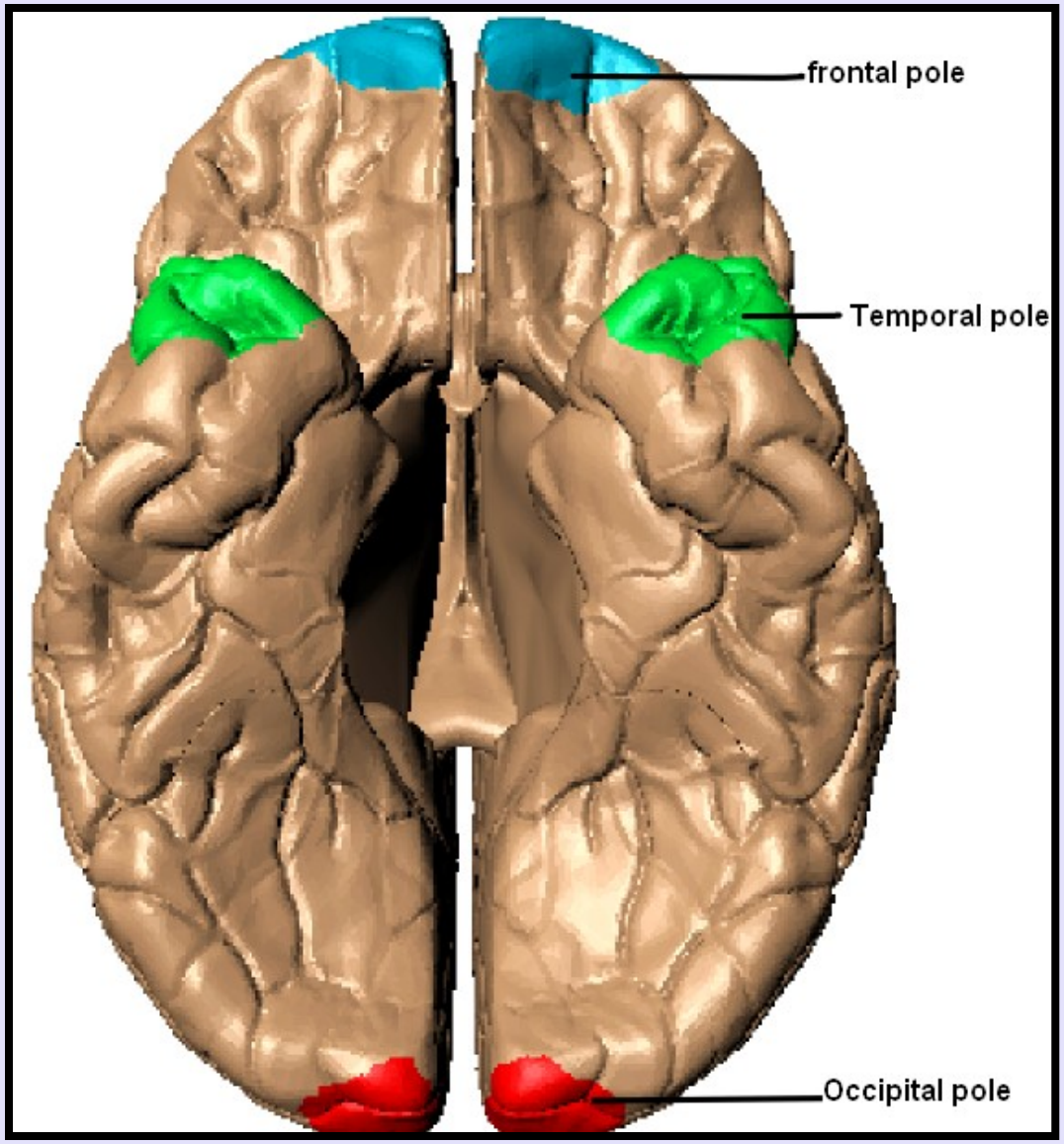
BORDERS OF HEMISPHERES

- 3 surfaces are separated by the following **borders:**
- (*a*) S**upero-medial,** between the lateral and medial surfaces.
- (*b*) I**nfero-lateral,** between the lateral and inferior surfaces; the anterior part of this border separating the lateral from the orbital surface, is known as the **superciliary border.**
- (*c*) M**edial occipital,** separating the medial and tentorial surfaces.
- (*d*) M**edial orbital,** separating the orbital from the medial surface



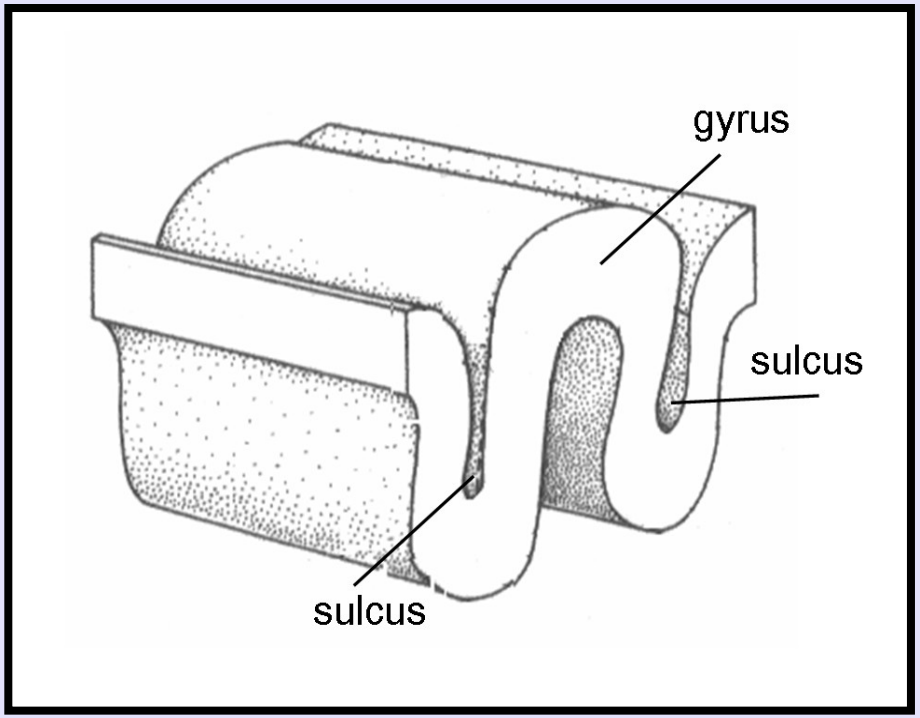
POLES OF CEREBRAL HEMISPHERES

- Frontal pole
  - The anterior end of the hemisphere
- Occipital pole
  - Posterior end of hemisphere
- Temporal pole
  - anterior end of the temporal lobe



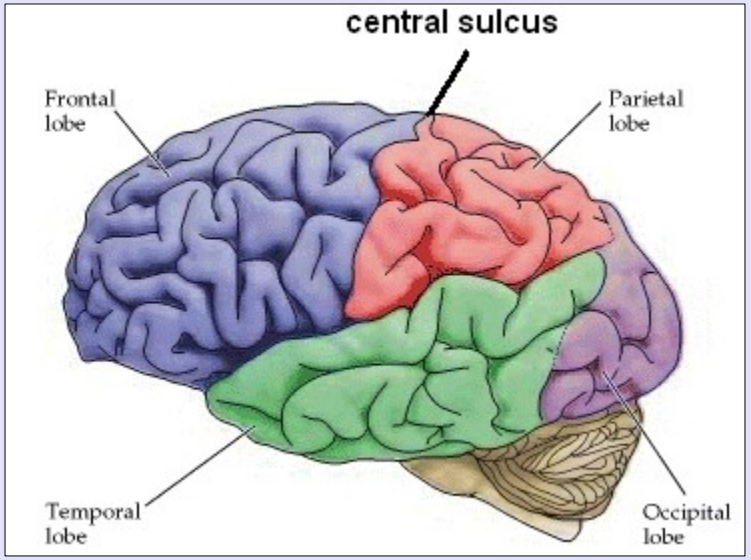
SULCUS AND GYRUS OF CEREBRAL HEMISPHERES

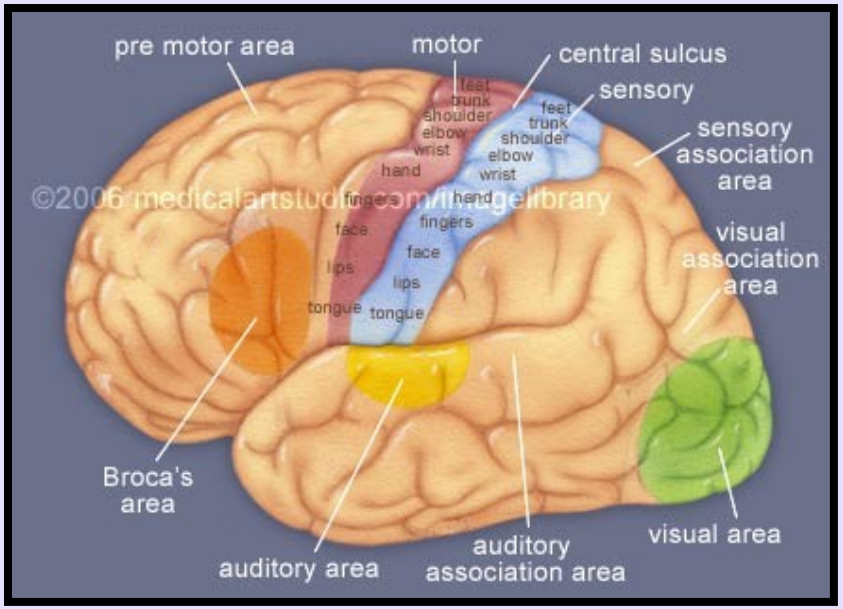
- Sulcus means "furrow“
  - Depression or fissure in the surface of the brain
  - Large furrows (sulci) that divide the brain into lobes are often called *fissures*.
- Gyrus ridge
  - Irregular eminences on the surface of brain formed by sulci
- The gyri and sulci are fairly constant in their arrangement



Interlobular sulci of cerebral hemispheres

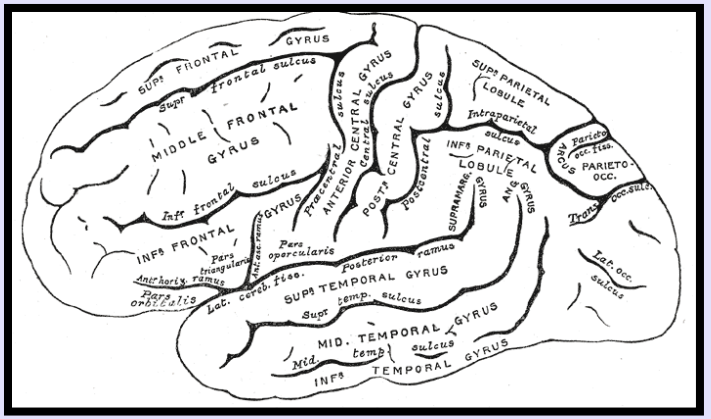
- Central sulcus
  - Middle of the lateral surface
  - Divides frontal and parietal lobe
  - Precentral gyrus contains motor area
  - Postcentral gyrus contain sensory area.

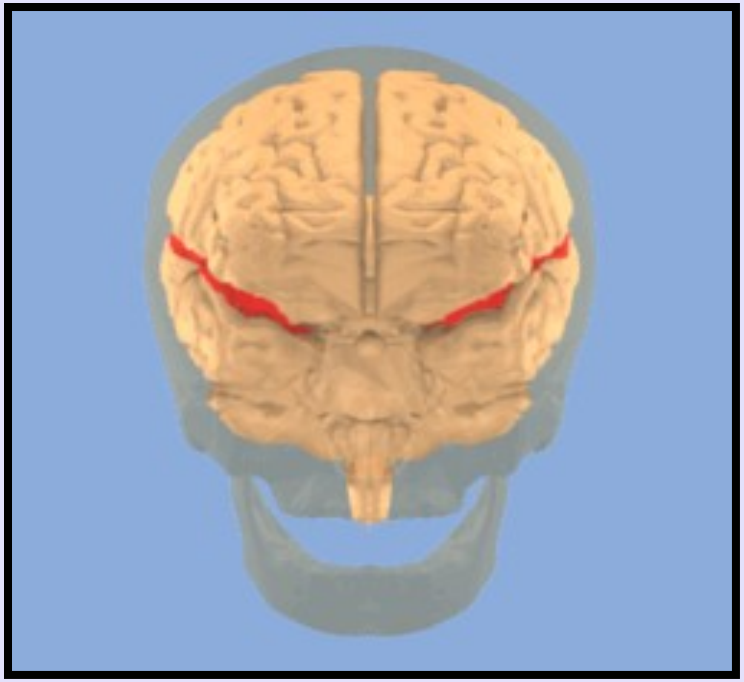




INTERLOBULAR SULCI OF CEREBRAL HEMISPHERES

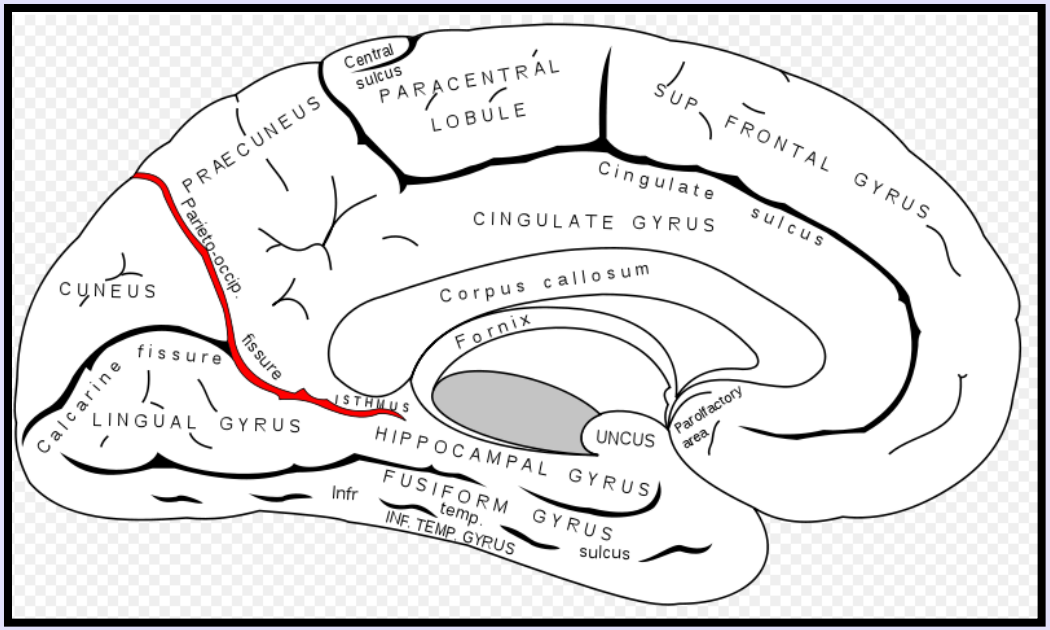
- Lateral sulcus
  - Divides frontal and parietal lobes above from temporal lobe
  - Divided into
  - Anterior horizontal ramus
  - Anterior ascending ramus
  - Posterior ramus

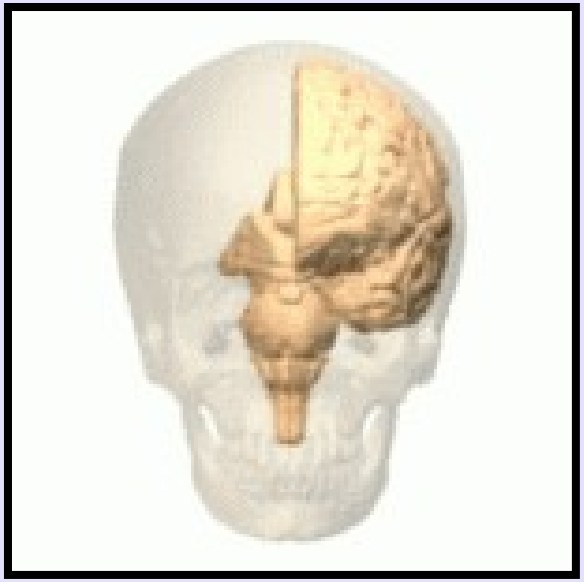




Interlobular sulci of cerebral hemispheres

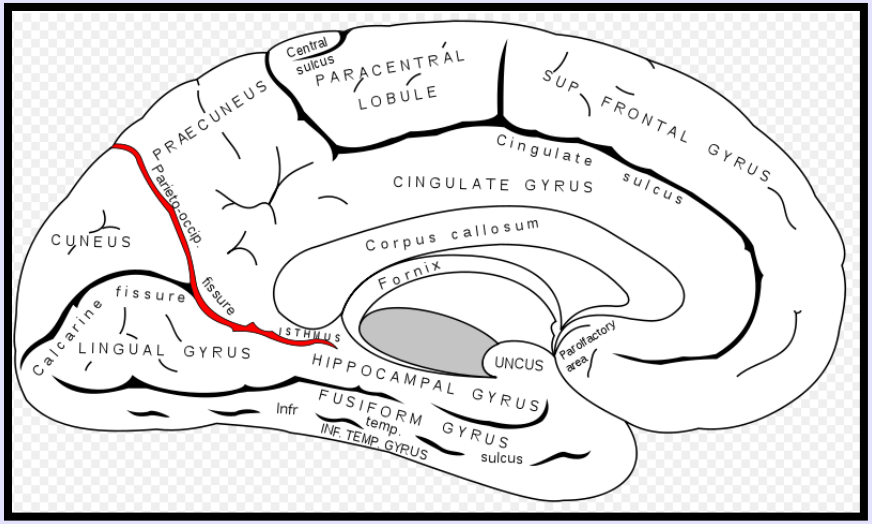
- Parieto-occipital sulcus
  - lateral surface = 5cm in front of occipital pole
  - Medial surface





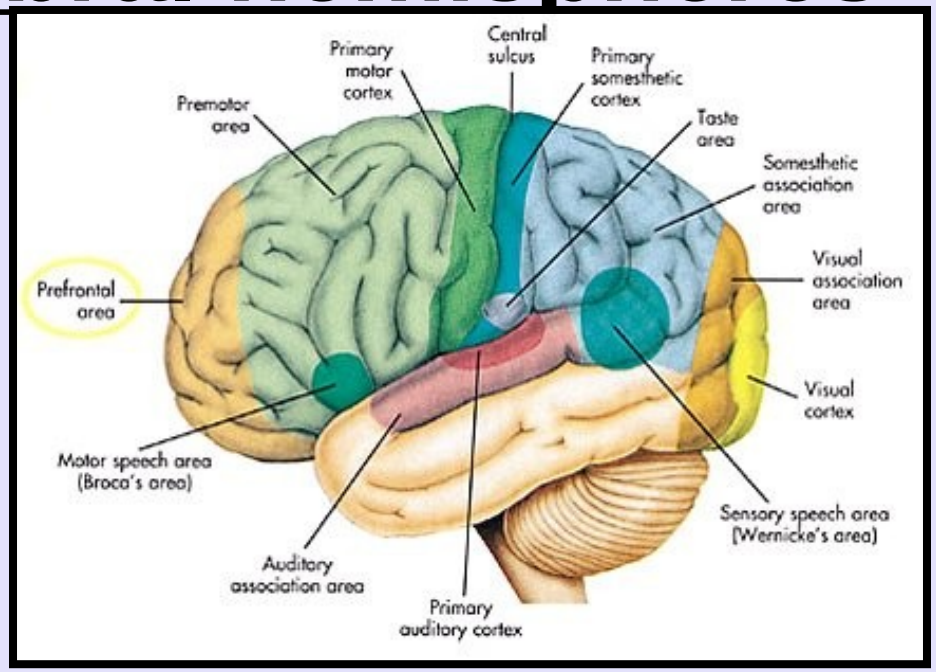
SULCI OF CEREBRAL HEMISPHERES

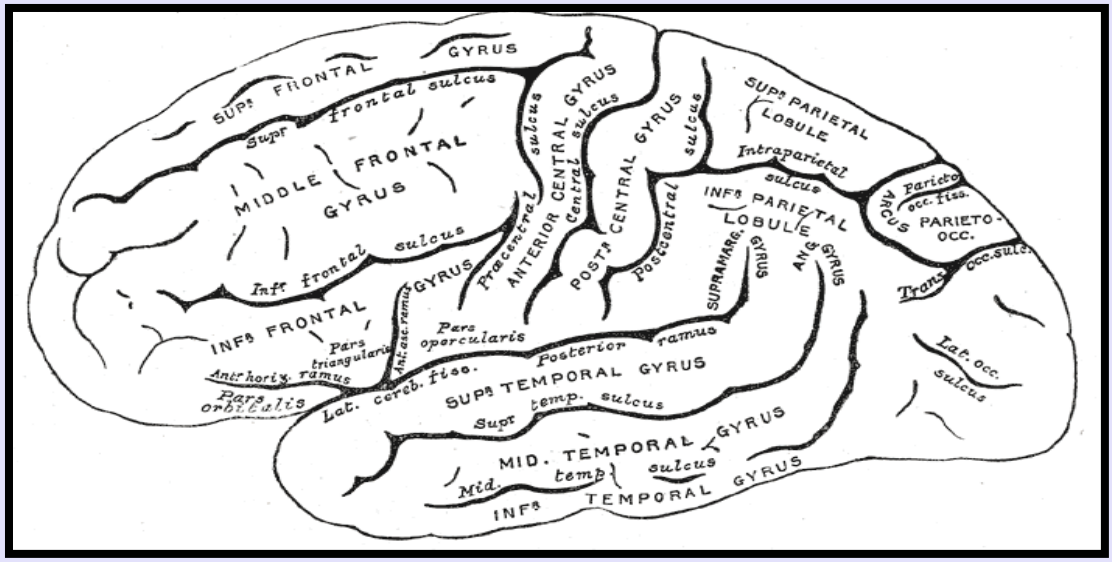
- Calcarine sulcus
  - Found on the medial surface
  - Starts behind corpus callosum
  - Arches back to the occipital pole



Lobes of cerebral hemispheres

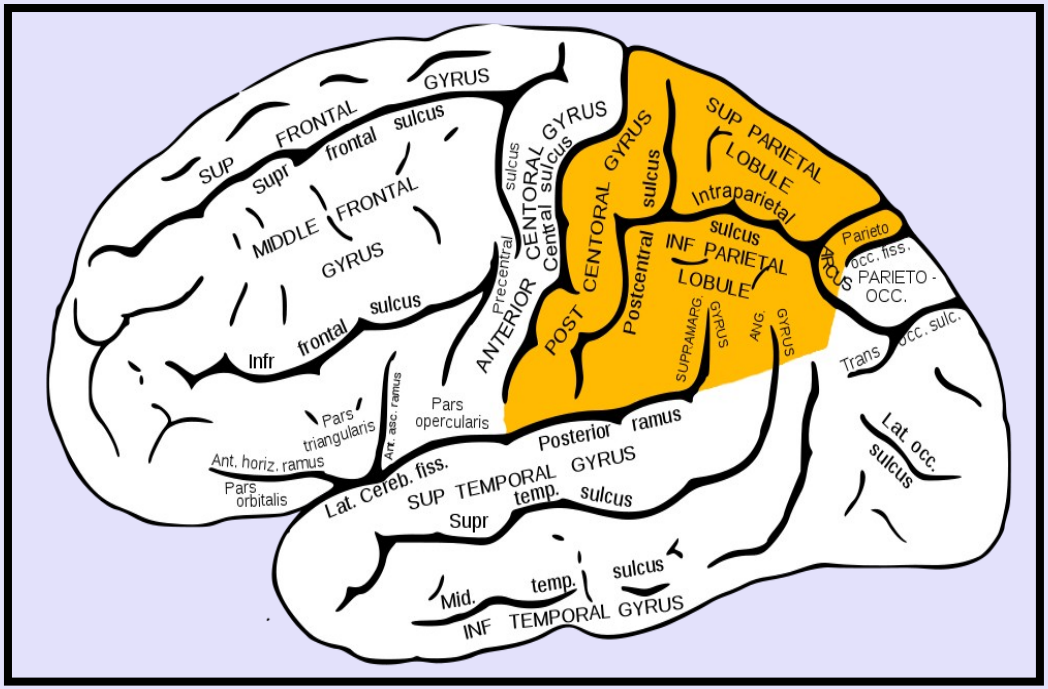
- **Frontal lobe**
- Three sulci divides it into four gyri
  - Precentral sulcus
  - Superior frontal sulcus
  - Inferior frontal sulcus
  - Superior frontal gyrus
  - Middle frontal gyrus
  - Inferior frontal gyrus
  - Precentral gyrus





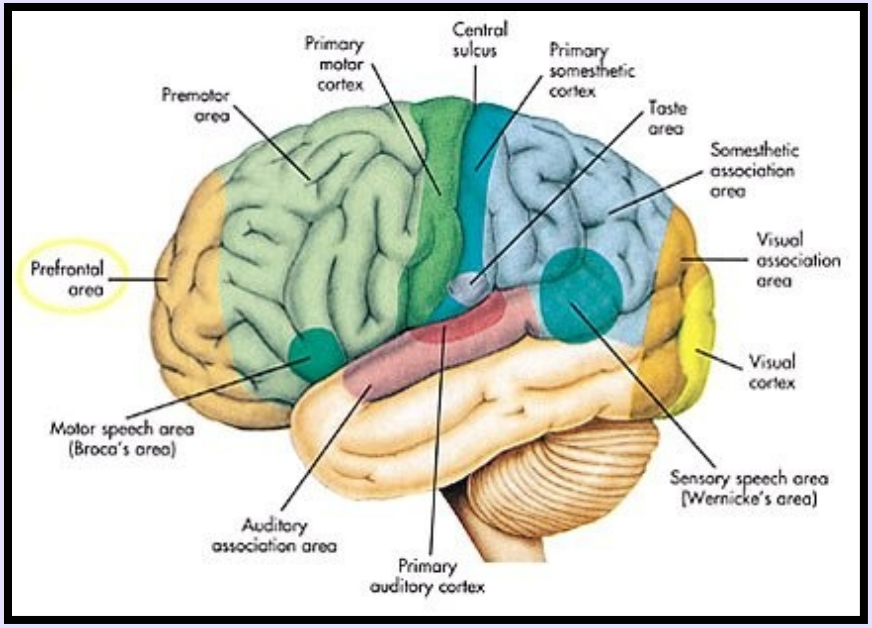
PARIETAL LOBE

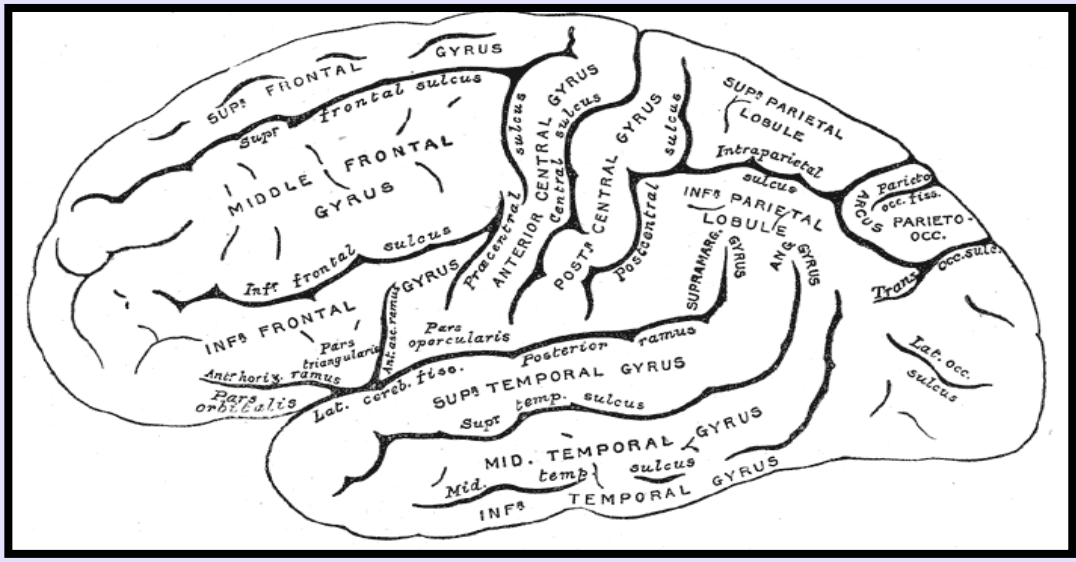
- 2 sulci divides into 3 gyri
  - Post central sulcus
  - Intraparietal sulcus
  - Post central gyrus
  - Supra parietal lobule
  - Infraparietal lobule
- Plays role in integrating sensory information from various parts of the body, knowledge of numbers and their relations, and in the manipulation of objects.



TEMPORAL LOBE

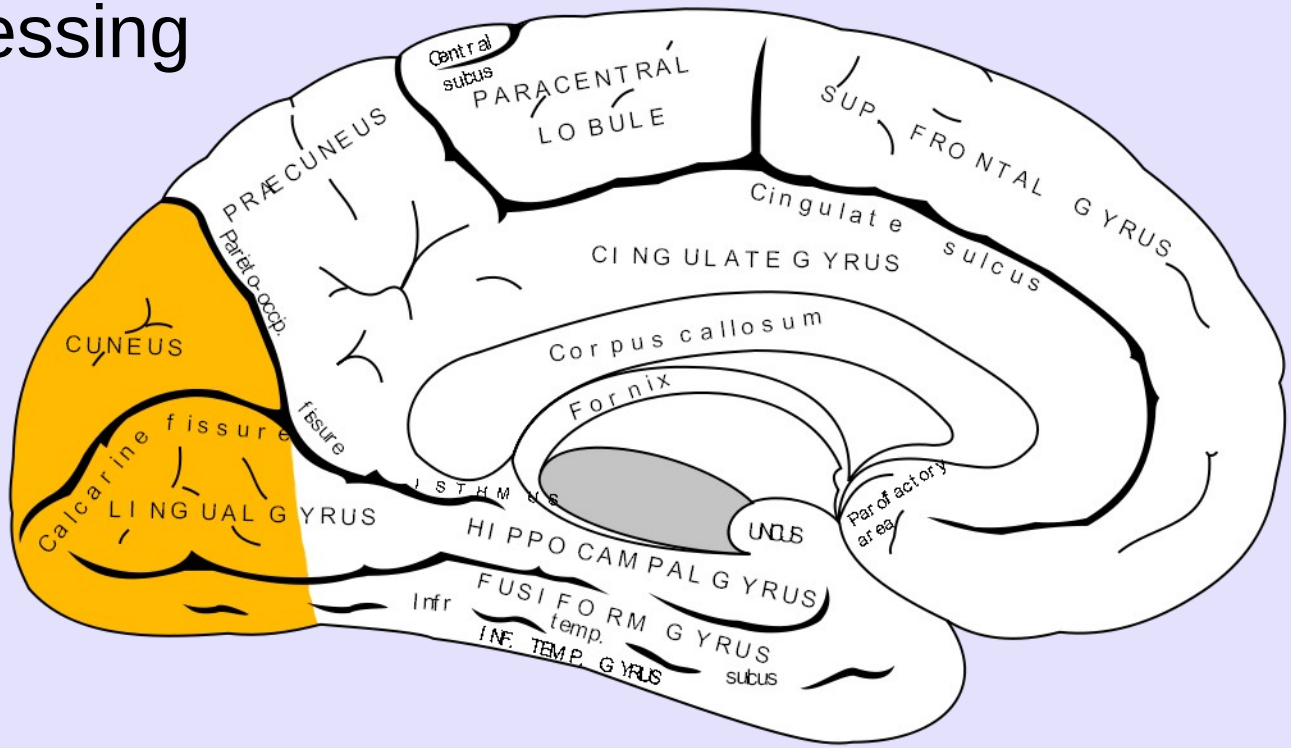
- 2 sulci divides it into 3 gyri
  - Superior temporal sulcus
  - Middle temporal sulcus
  - Superior temporal gyrus
  - Mid temporal gyrus
  - Inferior temporal gyrus
- The temporal lobe is involved in auditory perception and is home to the primary auditory cortex





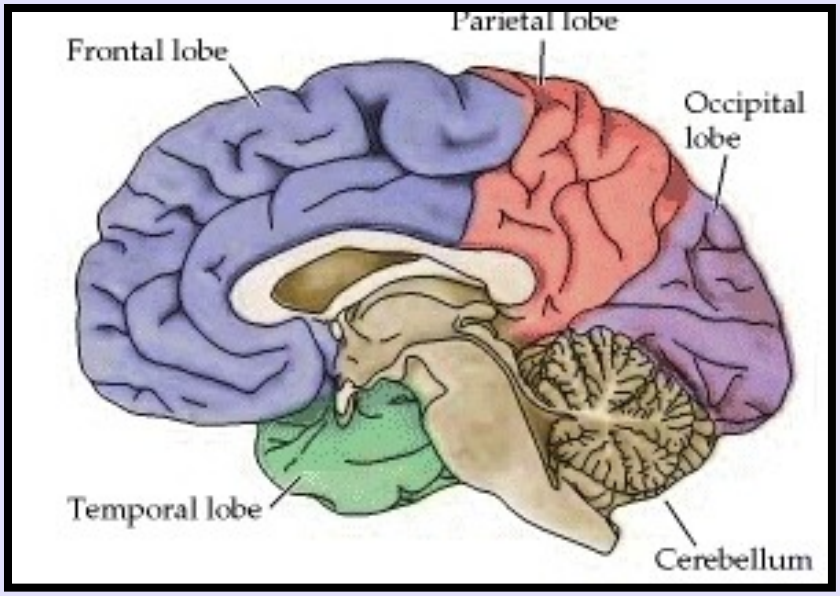
OCCIPITAL LOBE

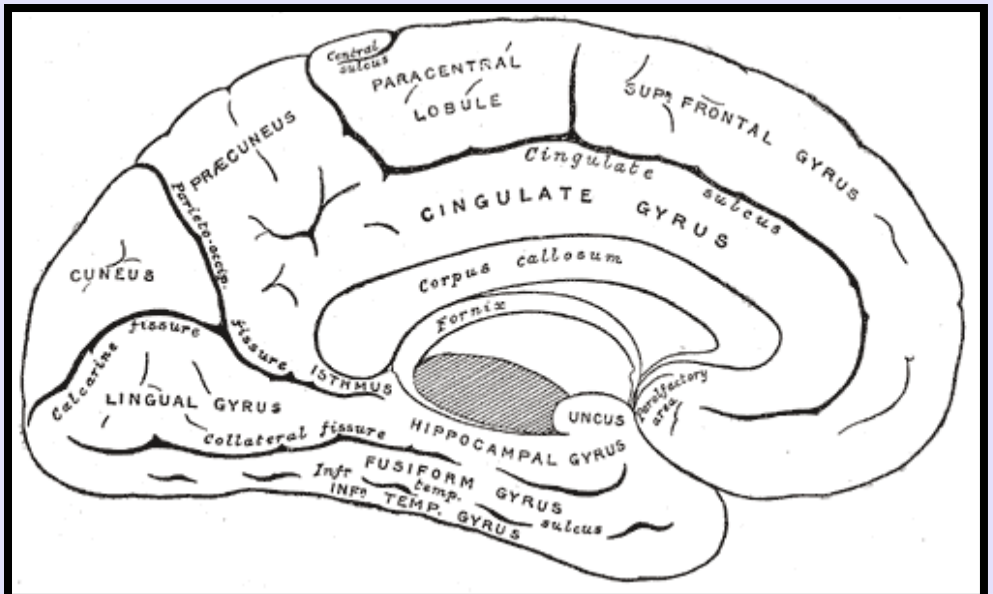
- Smallest lobe
- Lateral occipital gyri
- Lateral occipital sulci
- Visual processing cortex



MEDIAL AND INFERIOR SURFACE

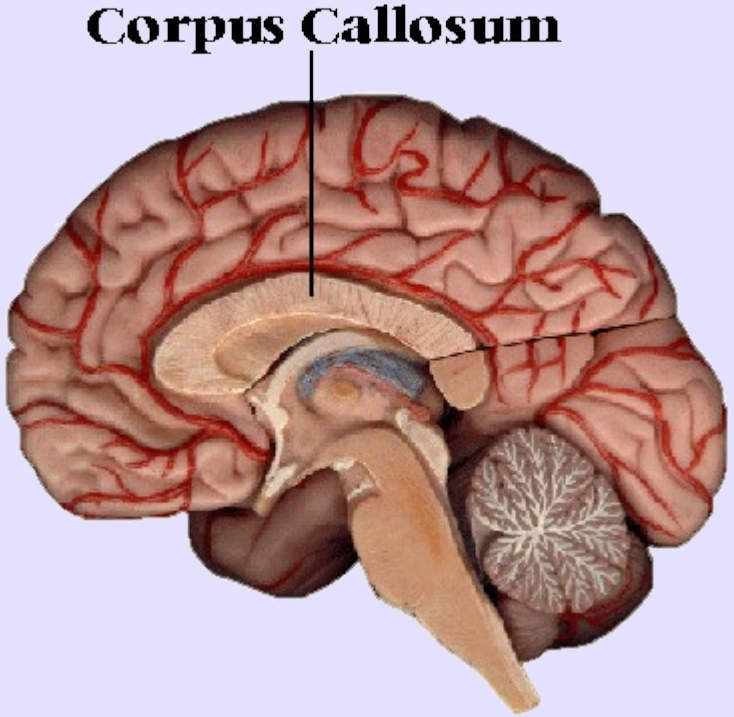
- Important structures on the medial surface
  - Corpus callosum
  - Cingulate gyrus
  - Callosal gyrus
  - Paracentral lobule
  - Precuneus
  - Cuneus
  - Collateral sulcus
  - Medial occipitotemporal gyrus

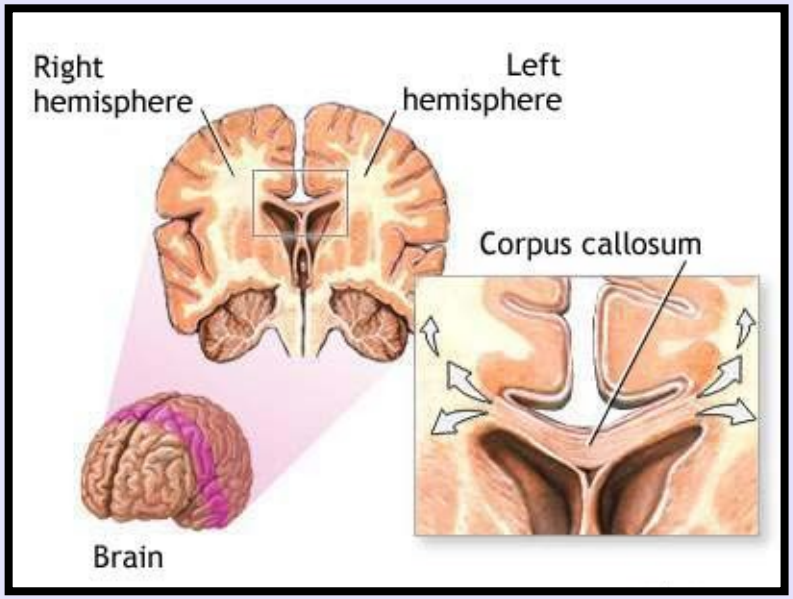




Corpus callosum

- Bundle of axons
- It connects the left and right cerebral hemispheres and facilitates interhemispheric communication

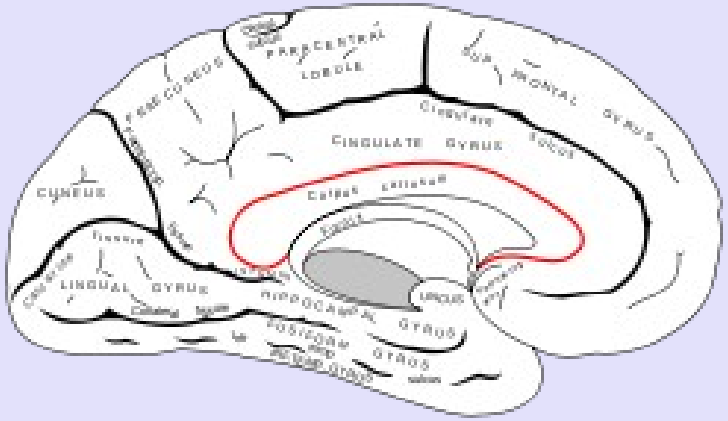




CINGULATE GYRUS

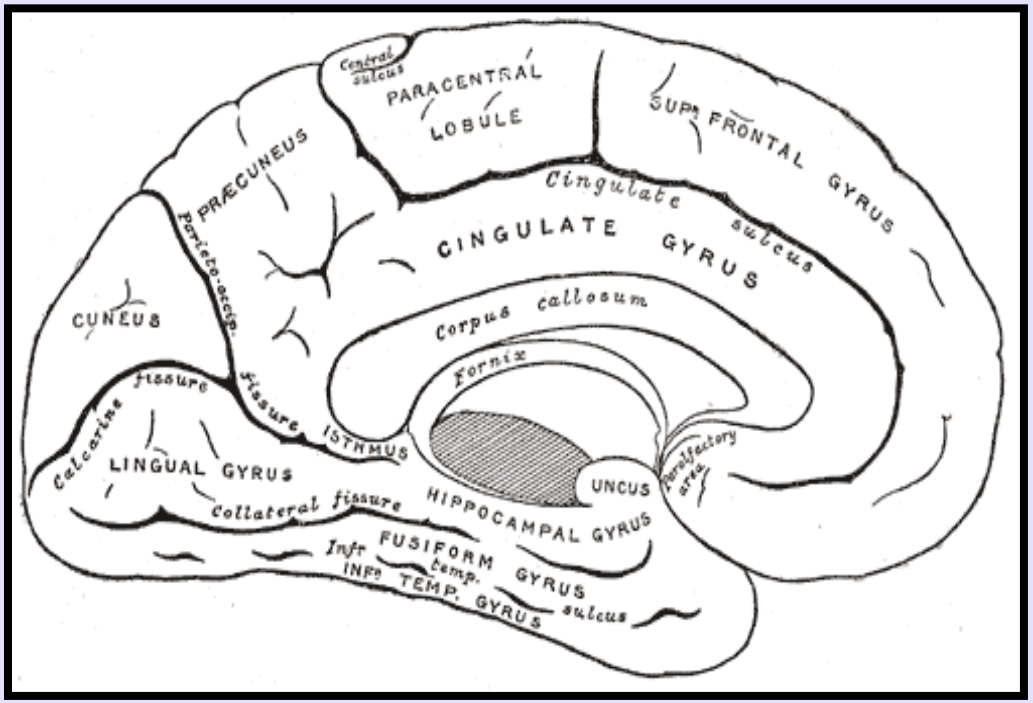
- Cingulate gyrus
  - Starts beneath the corpus callosum and goes back above it and ends at the posterior end of it.
- Callosal sulcus
  - Separates corpus callosum from cingulate gyrus
- Cingulate sulcus
  - Separates cingulate gyrus from superior frontal gyrus





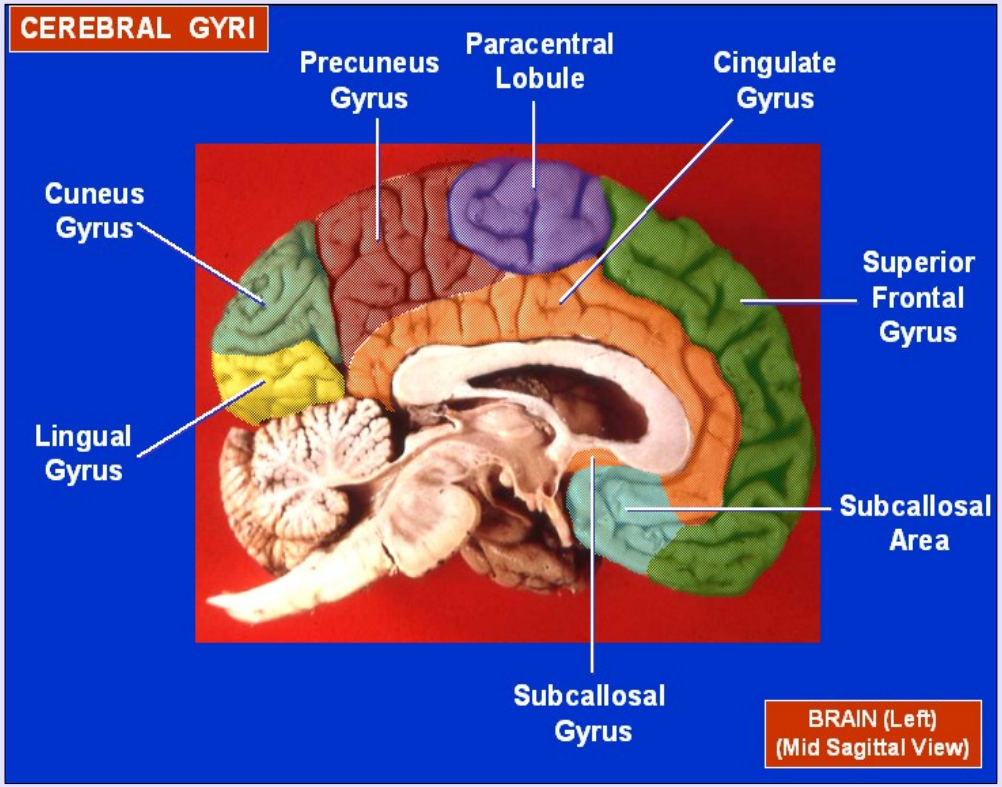
PARACENTRAL LOBULE

- Area of brain that surrounds the indentation formed by the central sulcus on the superior border
- Anterior part = precentral gyrus
- Posterior part = postcentral gyrus



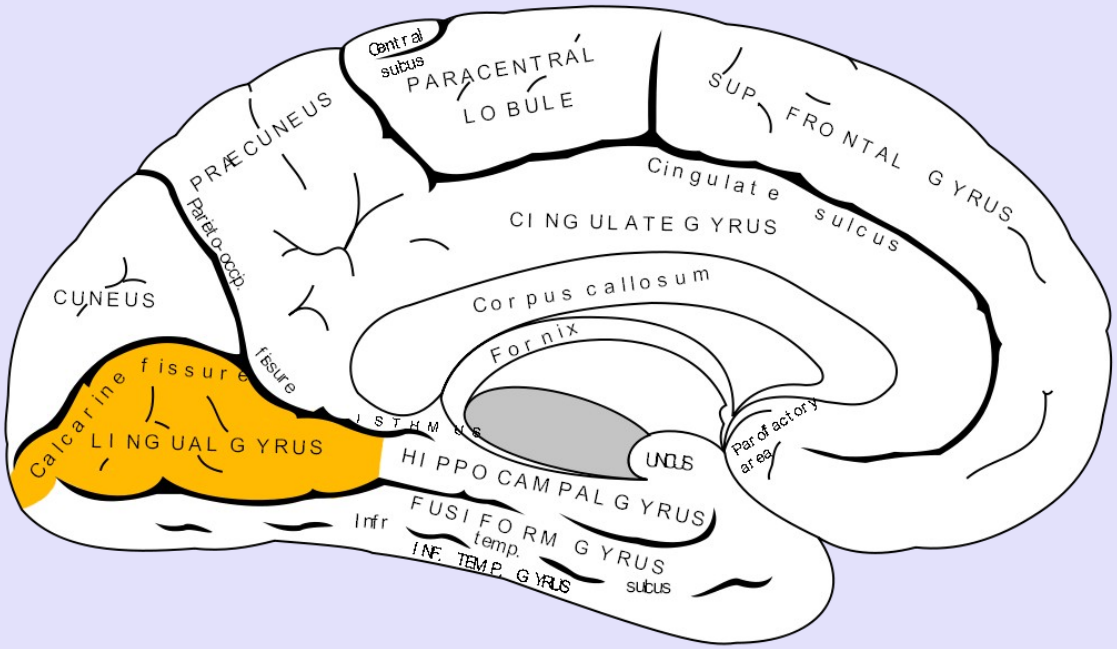
CUNEUS AND PRECUNEUS

- Precuneus
  - Anteriorly = upturned end of cingulate sulcus
  - Posterioly = parieto-occipital sulcus
- Cuneus
  - Triangular area between parieto-occipital sulcus and calcarine sulcus



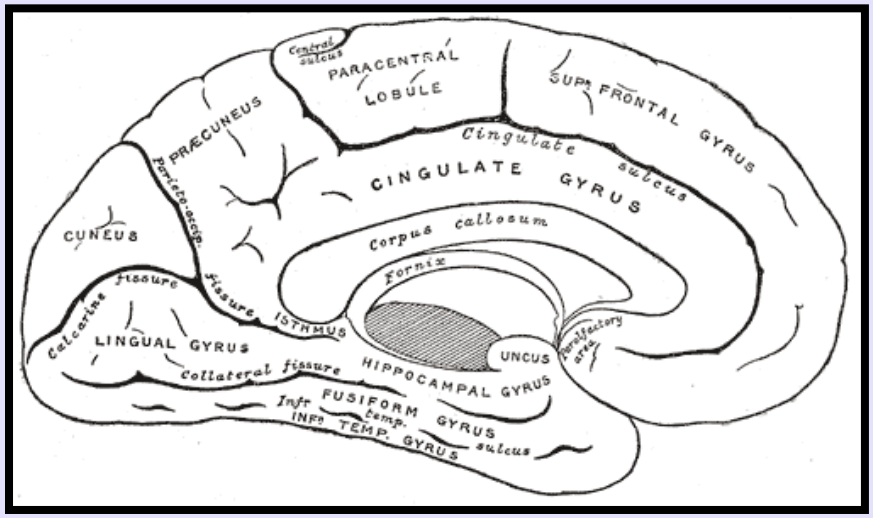
COLLATERAL SULCUS AND LINGUAL GYRUS

- On inferior surface
- Lingual gyrus
  - Between collateral sulcus and calcarine sulcus
- Parahippocampal gyrus
  - Anterior to lingual gyrus

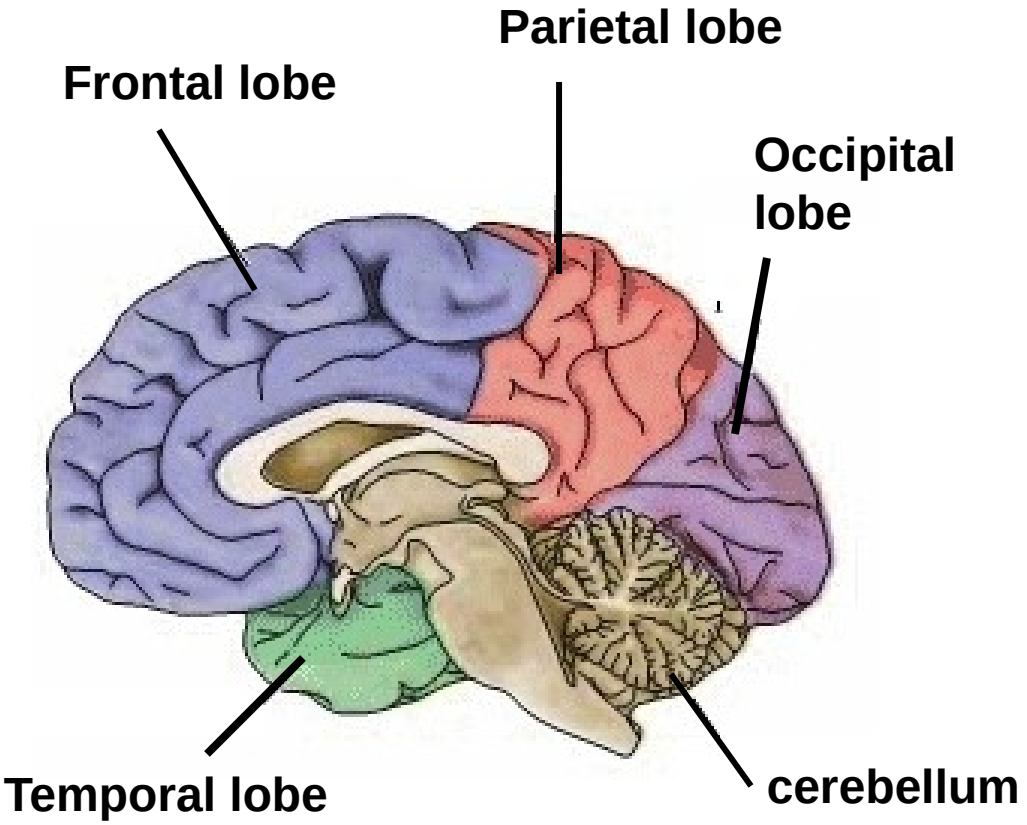


MEDIAL OCCIPITO-TEMPORAL GYRUS

- From occipital pole to temporal pole
  - Medially by collateral sulcus
  - Laterally by occipitotemporal sulcus
- Continuous with inferior temporal gyrus



Identify??


Parietal lobe
Frontal lobe
Occipital lobe
Temporal lobe
cerebellum

THANK YOU